

Ellen Lee

Education

University of Hawai'i at Manoa – Institute for Astronomy	Dec 2023 – Present
Ph.D. candidate in Astronomy	
Dissertation: Adaptive secondary mirror testing and development	
Advisors: Michael Connelley and Mark Chun	
University of Hawai'i at Manoa – Institute for Astronomy	Aug 2021 – Dec 2023
M.S. in Astronomy. GPA: 4.0/4.0.	
Cornell University	Aug 2017 – May 2021
B.S. with Honors in Engineering Physics. Magna cum laude. GPA: 3.78/4.0.	
Thesis: The FPI Metrology System for EoR-Spec	
Advisors: Gordon Stacey and Thomas Nikola	

Research Experience

Graduate Research Assistant	Aug 2021 – Present
University of Hawai'i at Manoa, Institute for Astronomy –NASA Infrared Telescope Facility	
Intern	Jun 2023 – Aug 2023
NASA Goddard Space Flight Center – Mars Sample Return CCRS Vision System lab	
Undergraduate Research Assistant	Jun 2018 – May 2021
Cornell University – Submillimeter Astrophysics Group	

Mentorship

Brianna Peck – Undergraduate at UC Berkeley	Summer 2025
Keilani Rapu – High school student	Fall 2024 – Spring 2025

Awards

ARCS Scholar of the Year	May 2026
Columbia Communications ARCS Award	Feb 2026
Center for Adaptive Optics Scholarship	Nov 2025
AAS International Travel Grant	Aug 2025
SPIE Student Travel Grant	Mar 2024
Shining Star Intern (NASA GSFC)	Aug 2023
Paul L. Hartman Prize in Experimental Physics	Jun 2021
Ng Yin Ying Scholarship	Aug 2019, Aug 2020
Friends of Cornell Astronomy Best Undergraduate Research Poster	May 2019
Roger and Mary Lou West Fellowship	Mar 2019

Observing Time Awarded

1. P.I., NASA IRTF, 4 hours, 26B. “An empirical timeline for planet formation through the evolution of protoplanetary disk morphology.”
2. P.I., Gemini North, 6.8 hours, 26B. “An empirical timeline for planet formation through the evolution of protoplanetary disk morphology.”
3. P.I., NASA IRTF, 93 hours, 25B-26A. “High resolution spectroscopy of a very young population of protostars.”
4. P.I., Gemini North, 24.4 hours, 25B-26A. “High resolution spectroscopy of a very young population of protostars.”
5. P.I., Subaru, 6.5 nights, 24A-24B. “Imaging disks around embedded protostars with SCEXAO.”

6. Co-I., NASA IRTF, 24A. “Measuring the Physical Stellar Parameters and Accretion Rates of Young Stars with Planet Forming Disks.”
7. Co-I., NASA IRTF, 24A-25B, “Rapid SpeX and ‘Ophi Follow-up of a New Kilonova.”

Selected Outreach Activities

Journey Through The Universe	Feb 2023 – Present
Maunakea Scholars	Sep 2021 – Present
AstroDay East and West	Oct 2022 – Present
Hawai’i State Science Olympiad	Mar 2022 – Mar 2024

Presentations

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1. SPIE, Adaptive Optics Systems X, Jul 2026. “Improving the throughput of facility instruments with IRTF-ASM-1.”
 2. Center for Adaptive Optics Science Retreat, Jan 2026. “Measuring non-common path aberrations with high dynamic range.”
 3. Center for Adaptive Optics Science Retreat, Jan 2026. “Progress report on the integration of the IRTF adaptive secondary mirror.”
 4. UC Santa Cruz - LAO Lunch talk, Jan 2026. “Facility integration of the IRTF adaptive secondary mirror.”
 5. University of Arizona - NOIRLab TLASH talk, Dec 2025. “Facility integration of the IRTF adaptive secondary mirror.”
 6. Adaptive Optics for Extremely Large Telescopes 8, Oct 2025, “Progress report on the integration of the IRTF adaptive secondary mirror.”
 7. Center for Adaptive Optics Fall Science Retreat, Nov 2024, “Status of the IRTF adaptive secondary mirror.”
 8. SPIE, Adaptive Optics Systems IX, Jul 2024, “First laboratory and on-sky results of an adaptive secondary mirror with TNO-style actuators on the NASA Infrared Telescope Facility.”
 9. CCAT-Prime/FYST Collaboration Meeting, 2021, part of “EoR Spectrograph.”
 10. Cornell Astronomy Undergraduate Research Forum, 2019, “Fabricating Spectral Filters and Creating the Calibration System for HIRMES.”

Peer-Reviewed Publications

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1. ***Lee, E.**, Flores, C., Connelley, M., & Williams, J., "Probing the ages of proplyds in the Orion Nebula from surface gravity measurements," *ApJ*. ***in prep**
 2. ***Lee, E.** & Chun, M., in prep, "Model-agnostic methods for efficiently measuring large non-common path aberrations on sky," *A&A*. ***in prep**
 3. **Lee, E.** 2026, "Sequential and nonsequential Zemax dynamic-link libraries for generating image slicer integral field units," *JATIS*, 12, 025003
 4. **Lee, E.** & Connelley, M. S. 2026, "Probing the kinematics of FU Orionis objects through high-resolution near-infrared spectroscopy," *ApJ*, 1004, 40
 5. Zhang, R., Lai, O., Peck, B., Chun, M., & **Lee, E.**, "CACOFONI: Cosine Amplitude Calibration Organized in Frequency for On-sky Nimble Interaction-matrix. A novel method for calibrating adaptive optics systems II," *AJ*, *in review*.
 6. Bos, B. J., et al., including **Lee, E.** 2024, "Vision System for the Mars Sample Return Capture Containment and Return System (CCRS)," *Aerospace*, 11, 456
 7. Kuhn, M. A., Hillenbrand, L. A., Connelley, M. S., et al., including **Lee, E.** 2024, "The 2022–2023 accretion outburst of the young star V1741 Sgr," *MNRAS*, 529, 2630

1. **Lee, E.**, Chun, M., Zhang, R., et al. 2026, "Enhancing the throughput of facility instruments with IRTF-ASM-1," in *Adaptive Optics Systems X, SPIE*
2. **Lee, E.**, Chun, M., Zhang, R., et al. 2026, "Progress report on the integration of the IRTF adaptive secondary mirror," in *Adaptive Optics for Extremely Large Telescopes 8 (AO4ELT8)*, Viña del Mar
3. **Lee, E.**, Chun, M., Lai, O., et al. 2024, "First laboratory and on-sky results of an adaptive secondary mirror with TNO-style actuators on the NASA Infrared Telescope Facility," in *Adaptive Optics Systems IX, SPIE*, 13097, 652
4. **Lee, E.**, Emerson, K. S., Connelley, M. S., et al. 2022, "Commissioning 'Opihi: a wide-angle finderscope for the NASA Infrared Telescope Facility," in *Ground-based and Airborne Instrumentation for Astronomy IX, SPIE*, 12184, 2710
5. Kuiper, S., Kidron, M., Boot, K., et al., including **Lee, E.** 2026, "Integration and Acceptance Testing of the UH2.2 Adaptive Secondary Mirror," in *Adaptive Optics Systems X, SPIE*
6. Zhang, R., Baeten, M., Chun, M. R., et al., including **Lee, E.** 2024, "In-lab and On-sky Closed-loop Results of Adaptive Secondary Mirrors with TNO's Hybrid Variable Reluctance Actuators," in *Adaptive Optics Systems IX, SPIE*, 13097, 1321
7. Bos, A., Dekker, B., Kuiper, S., et al., including **Lee, E.** 2024, "Status and first results of the NASA IRTF Adaptive Secondary Mirror," in *Advances in Optical and Mechanical Technologies for Telescopes and Instrumentation VI, SPIE*, 13100, 271